CLOSURES OF DISJOINT OPEN SUBSETS OF AN EXTREMALLY DISCONNECTED SPACE ARE DISJOINT

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *Let X be an extremally disconnected space. If $U, V \subset X$ are disjoint open subsets, then $\overline{U}$ and $\overline{V}$ are disjoint too.*

Proof. By assumption $\overline{U}$ is open, hence $V \cap \overline{U}$ is open and disjoint from U, hence empty because $\overline{U}$ is the intersection of all the closed subsets of X containing U. This means the open $\overline{V} \cap \overline{U}$ avoids V hence is empty by the same argument. $\square$

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